

Explainable Machine Learning for Multi-Class Power Quality Disturbance Classification Using SHAP and Feature Importance Analysis

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Abstract—The reliability and efficiency of modern power systems are negatively impacted by power quality disturbances (PQDs), such as voltage sags, swells, harmonics, and transients. Hence, rapid and accurate classification of PQDs is essential to prevent equipment failures and to ensure stable grid operation. This paper proposes an explainable machine learning (ML) framework for multi-class PQD classification. The framework combines traditional algorithms, Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), Support Vector Machine (SVM), and XGBoost, with a one-dimensional Convolutional Neural Network (1D-CNN). The dataset was preprocessed by handling negative THD values, capping outliers, performing feature selection, encoding categorical variables, and normalizing numerical features. Model interpretability was ensured using feature importance analysis and SHAP values. The experimental findings indicate that LR achieved the maximum accuracy of 99.79% across the five disturbance categories. Beyond predictive performance, the framework enhances transparency by identifying the most influential features including Total Harmonic Distortion (THD), RMS Voltage, and Peak Voltage. The findings demonstrate that integrating robust preprocessing, high-performing models, and explainability methods offers a reliable and interpretable solution for detecting power disturbances in advanced grid networks and present-day energy infrastructures.

Keywords—Power quality disturbances, Power quality monitoring, Machine learning, Explainable AI (XAI), SHAP, 1D-CNN, Smart grids

I. INTRODUCTION

The global shift toward distributed and renewable energy generation has intensified the occurrence and variety of PQDs, such as voltage sags, swells, harmonics, and transients, posing significant risks to equipment performance and grid stability [4], [15]. Undetected or misclassified PQDs can degrade performance, trigger nuisance trips, shorten asset lifespan, and cause economic losses for utilities and consumers [8], [11]. Traditional PQD analysis using transform-based techniques (Fourier, wavelet, S-transform) and rule-based detectors, though effective in some cases, often requires expert interpretation and struggles with nonstationary or hybrid disturbances [6], [9]. To overcome these limitations, ML approaches have been widely adopted. Early studies employed kernel-based models and one-versus-one SVMs for multiclass PQD classification [1], [2]. More recently, CNNs, deep belief networks, and hybrid models have excelled by automatically

extracting temporal and spectral features [3], [5]. Feature engineering techniques, such as empirical wavelet transforms, variational mode decomposition, and metaheuristic optimization, produced compact, discriminative inputs [6], [12], while ensemble methods and refined SVMs enhanced accuracy and robustness [10], [13]. Visualization approaches, like space vector ellipse representations, aid in distinguishing similar disturbance classes [4], [8]. However, challenges remain: limited interpretability and robustness in noisy environments. Many high-performing ML and deep learning (DL) models act as “black boxes,” providing minimal physics-aligned explanations [6], [12], and practical deployment requires rigorous preprocessing, outlier handling, and validation [7], [11]. Overall, trustworthy PQD monitoring demands robust preprocessing, effective feature engineering, powerful classifiers, and explainability [3], [14].

Motivated by these needs, this paper proposes an explainable ML-based framework for multi-class PQD classification. The framework integrates systematic preprocessing, diverse classifiers including LR, ensemble methods, and a 1D-CNN, and interpretability analysis using feature-importance metrics and SHAP values.

This study makes the following key contributions:

- 1) We introduce an ML-based system for multi-class PQD classification that integrates six diverse classifiers with a comprehensive preprocessing pipeline.
- 2) We demonstrate that LR achieved the highest accuracy (99.79%), closely followed by XGBoost and a 1D-CNN (99.58%), establishing a strong performance benchmark across five disturbance classes.
- 3) By combining SHAP and feature-importance analysis, we identify and validate critical features such as THD, RMS voltage, and Peak Voltage, bridging the gap between data-driven predictions and engineering insight.

The rest of this paper is arranged as follows: Section II analyzes prior work in the field of ML-based classification of PQD. Section III presents the dataset, exploratory data analysis (EDA), preprocessing methods, and experimental framework. The results and analysis are discussed in Section IV, while Section V emphasizes the conclusions and articulates subse-

quent research directions.

II. RELATED WORK

The proper classification of PQDs is vital for maintaining power system stability amidst increasing renewable integration. Research has evolved from traditional signal processing to advanced computational intelligence. Early studies, such as Bhagat et al. [21], combined wavelet transforms with artificial neural networks, though these methods required manual feature engineering and struggled with complex disturbances. The emergence of DL has facilitated end-to-end classification; for instance, Ekici et al. [18] employed Bayesian CNNs, whereas Sindi et al. [16] introduced a hybrid approach that combines 1D-CNNs for processing raw signals with 2D-CNNs for analyzing time–frequency representations. Recurrent architectures were also explored: Yiğit et al. [20] combined GRUs with CNNs, while Khetarpal et al. [19] used a bidirectional LSTM with dual attention to capture long-range temporal patterns. Concurrently, feature optimization gained attention, with Han et al. [17] using CNNs for atomic decomposition, Chamchuen et al. [23] applying an adaptive salp swarm optimizer, and Zhang et al. [22] employing visual-attention to highlight key signal portions.

Although these advances have improved predictive accuracy, a major limitation remains: Most state-of-the-art models act as ‘black boxes,’ providing minimal clarity into how decisions are made. This lack of interpretability is a critical barrier to practical deployment in power systems, where operators and engineers require physically meaningful explanations to validate and trust automated outputs. Addressing this explainability gap is the central motivation of the present study.

III. METHODOLOGY

This section describes the proposed explainable ML framework, which includes EDA, preprocessing, model training, evaluation, and interpretability, as illustrated in Fig. 1.

A. Dataset Overview

This study employed a publicly accessible dataset obtained from Kaggle [24] and comprises labeled signals for five PQD classes: Normal, Sag, Swell, Harmonics, and Transient. Each record includes electrical features capturing time- and frequency-domain characteristics, such as RMS voltage, peak voltage, and total harmonic distortion (THD). To better represent transient and non-stationary patterns, discrete wavelet transform (DWT)-based features were also incorporated. A categorical variable representing the signal phase was numerically encoded for ML compatibility. The dataset is balanced across all classes, ensuring unbiased training and evaluation of the models.

B. Exploratory Data Analysis (EDA)

An EDA was performed to examine dataset characteristics, including summary statistics, target class distribution, and feature correlations.

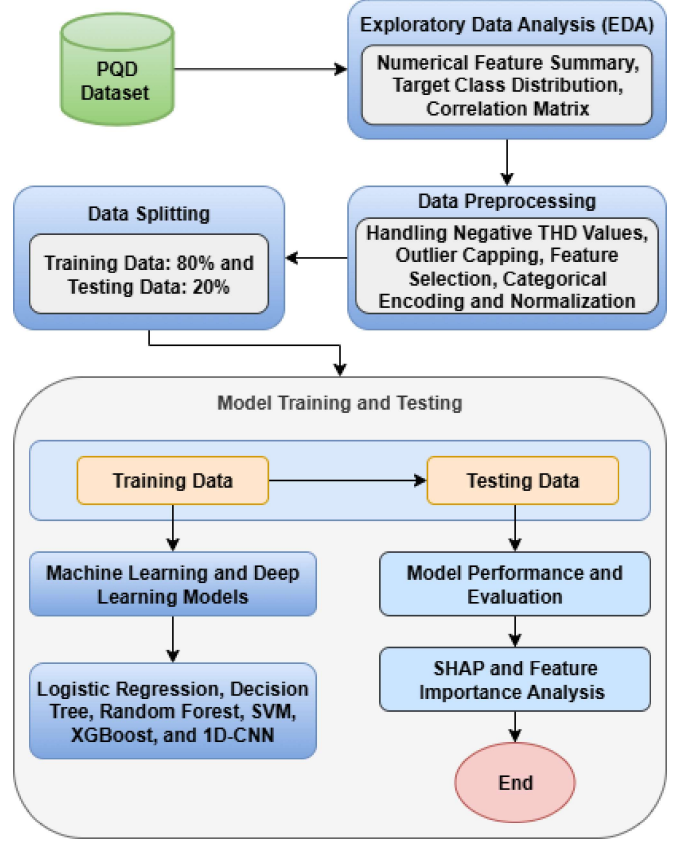


Fig. 1. Proposed framework for PQD classification.

- (a) *Numerical Feature Summary*: The dataset contains 2367 labeled records, described by nine numerical features capturing time-domain and frequency-domain characteristics. Table I summarizes these numerical features.

TABLE I
SUMMARY STATISTICS OF NUMERIC FEATURES.

Feature	Count	Mean	Std	Min	Max
ID	2367	1184.00	683.44	1.00	2367.00
RMS_Voltage (V)	2367	221.91	39.64	113.99	300.93
Peak_Voltage (V)	2367	328.81	66.78	161.21	465.68
THD (%)	2367	4.74	3.33	-0.25	18.94
Duration_ms	2367	153.10	82.79	10.00	299.00
DWT_Energy_Level1	2367	54.34	25.87	10.00	99.95
DWT_Energy_Level2	2367	27.49	12.94	5.01	49.99
DWT_Entropy	2367	3.03	1.16	1.00	5.00
Signal_Noise_Ratio_dB	2367	30.00	5.70	20.01	39.99
Label	2367	1.996	1.43	0.00	4.00

- (b) *Class Distribution of Target Variable*: The label, which serves as the target variable, represents five classes of PQDs: Normal, Sag, Swell, Harmonics, and Transient. The dataset is approximately balanced, with each class containing between 450 and 500 samples. The class distribution is illustrated in Fig. 2.
- (c) *Correlation Matrix*: To evaluate relationships among numerical features, Pearson correlation coefficients were

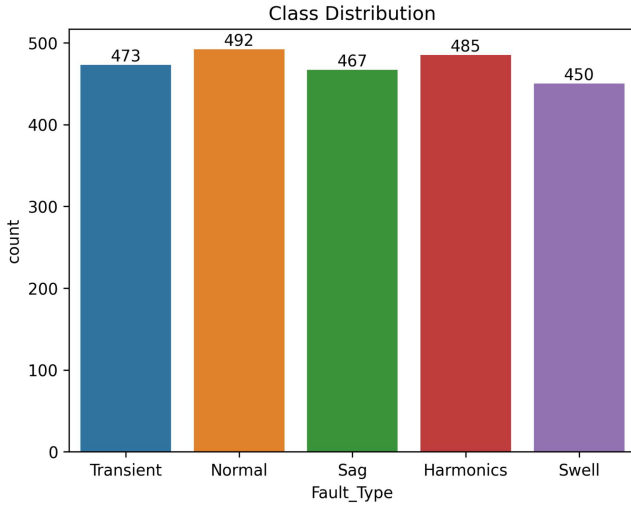


Fig. 2. Distribution of records across the five PQD classes.

computed. The correlation heatmap in Fig. 3 indicates strong correlations between RMS Voltage and Peak Voltage and between THD and the target label, while other features provide complementary information for classification.

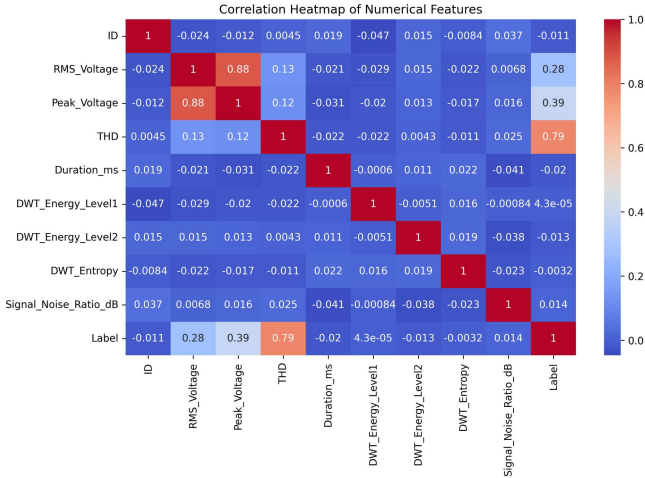


Fig. 3. Correlation heatmap of numerical features.

C. Data Preprocessing

The raw dataset was preprocessed to ensure physical validity and statistical consistency before training the classification models. The following steps were applied.

- (a) *Handling Invalid THD Values:* Since THD is always a non-negative measure of harmonic distortion in electrical signals, negative values observed in the dataset were considered invalid. These values were replaced with zero to maintain the physical correctness of the feature space.

- (b) *Outlier Capping:* Outliers were detected using the interquartile range (IQR) method, where values not contained within

$$[Q_1 - 1.5 \times IQR, Q_3 + 1.5 \times IQR] \quad (1)$$

were considered extreme. RMS Voltage exhibited 868 outliers, followed by Peak Voltage with 325 and THD with 192, while no outliers were observed in other features. In total, 1385 outlier instances were identified with overlaps. These values were capped at their lower or upper bounds to reduce noise while preserving the overall distribution.

- (c) *Feature Selection:* Non-informative attributes such as ID, Fault_Type, and Label were excluded to prevent data leakage, ensuring that the model learns to make predictions using only physically meaningful input features. The remaining features including RMS Voltage, Peak Voltage, THD, DWT-based features, signal-to-noise ratio, duration, and encoded phase were retained as predictive variables for classifying the five categories of PQDs.
- (d) *Categorical Encoding:* The *Phase* feature, being categorical, was transformed into a numeric format via label encoding to align with the requirements of ML algorithms.
- (e) *Normalization:* Z-score standardization was applied to continuous numerical features, re-scaling them to a zero mean and unit standard deviation. This ensures equal weighting of all features during training and avoids bias toward attributes with larger numerical ranges.

D. Data Splitting

After preprocessing, the data was split 80:20 into training and testing sets using stratified sampling to preserve class distribution, with training for model learning and testing for evaluation.

E. ML Models

To establish baseline performance, several traditional ML models were trained and evaluated.

Logistic Regression (LR): LR is a simple, yet effective linear classifier commonly used for multiclass classification problems. Its interpretability and ease of implementation make it a standard baseline in data analysis.

Decision Tree (DT): A DT represents a non-linear approach that relies on a branching structure to split data by feature values and derive predictions. It was chosen due to its ease of interpretation and its ability to efficiently manage both categorical and numerical data types with limited preprocessing needs.

Random Forest (RF): RF operates as an ensemble of DTs whose predictions are combined to increase accuracy and mitigate the risk of overfitting. It was chosen to ensure more robust and widely applicable performance relative to a standalone DT.

Support Vector Machine (SVM): SVM separates data using an optimal hyperplane and, with kernel functions, can

handle nonlinear boundaries, making it effective for high-dimensional datasets.

XGBoost: XGBoost is a gradient boosting ensemble technique that builds sequential trees to minimize classification errors. This model was selected because of its strong performance, scalability, and ability to identify complex patterns in the data.

The hyperparameters of the baseline ML models are summarized in Table II.

TABLE II
HYPERPARAMETERS OF CLASSICAL ML MODELS

Model	Hyperparameters / Details
LR	max_iter = 1000
DT	default parameters
RF	n_estimators = 100
SVM	kernel = rbf, probability = True
XGBoost	eval_metric = mlogloss

1D Convolutional Neural Network (1D-CNN): A 1D-CNN applies convolutional filters along the feature dimension to learn hierarchical patterns from ordered PQD features. It captures local dependencies effectively while remaining lightweight for small to medium datasets. The architecture and hyperparameters are summarized in Table III.

TABLE III
1D-CNN ARCHITECTURE AND HYPERPARAMETERS

Layer Type	Parameters / Details
Conv1D Layer 1	32 filters, kernel size = 2, ReLU
Dropout	rate = 0.2
Conv1D Layer 2	64 filters, kernel size = 2, ReLU
Flatten	—
Dense Layer	128 neurons, ReLU
Output Layer	Softmax, number of classes = 5
Optimizer	Adam
Epochs / Batch	20 / 32

F. Model Evaluation

The trained ML and DL models were assessed using a range of performance metrics to provide a comprehensive evaluation.

- (a) *Accuracy:* Accuracy indicates how often the model's predictions are correct compared to the actual outcomes. It can be formulated as in (2).

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Samples}} \quad (2)$$

- (b) *Precision:* Precision calculates the percentage of predicted positives that are actually correct. It is defined as in (3).

$$\text{Precision} = \frac{\text{Correct Positive Predictions}}{\text{All Predicted Positives}} \quad (3)$$

- (c) *Recall:* Recall represents the fraction of true positive cases that the model accurately detects. It can be written as in (4).

$$\text{Recall} = \frac{\text{Correct Positive Predictions}}{\text{All Actual Positives}} \quad (4)$$

- (d) *F1-Score:* The F1-score offers a single metric that integrates Precision and Recall using their harmonic mean. It can be mathematically defined as in (5).

$$F1 = \frac{2 \cdot PR}{P + R} \quad (5)$$

Here, P represents Precision and R represents Recall.

- (e) *Confusion Matrix:* The confusion matrix reveals misclassifications by comparing predictions with true labels.

For the 1D-CNN, categorical cross-entropy loss and accuracy were tracked during training, with final performance evaluated on the unseen test dataset. This allowed direct comparison with traditional ML models to identify the most effective approach.

All model development and evaluation procedures were carried out using Python within a Kaggle Notebook, which supplied the essential computational environment and libraries.

IV. RESULTS AND DISCUSSION

This section compares model performance using selected metrics and examines feature importance and SHAP analyses to identify the most influential predictors.

A. Overall Classification Performance

Table IV summarizes the classification performance of all models. LR achieved the highest accuracy of 99.79%, followed by XGBoost and 1D-CNN at 99.58%, while DT, RF, and SVM also exceeded 99%, demonstrating that both classical and DL approaches perform reliably with robust preprocessing.

TABLE IV
PERFORMANCE EVALUATION ACROSS ML AND DL MODELS.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
LR	99.79	99.80	99.79	99.80
DT	99.37	99.38	99.37	99.37
RF	99.37	99.38	99.37	99.37
SVM	99.16	99.22	99.14	99.17
XGBoost	99.58	99.58	99.59	99.58
1D-CNN	99.58	99.60	99.58	99.59

To further illustrate the performance of the best model, the classification report of LR is shown in Fig. 4. The model achieved near-perfect scores across all five PQD classes, with precision, recall, and F1-score values consistently at or very close to 1.00. Class 0 (Normal) had a recall of 1.00, while Class 4 (Harmonics) recorded a slightly lower recall of 0.99, indicating minor misclassification. The weighted average across all classes confirms that the model performs uniformly well, without favoring any specific class. LR's superior performance is due to the linear separability of PQ disturbance features, allowing it to achieve high accuracy on smaller datasets, unlike deeper models like 1D-CNN that require larger data to capture complex patterns.

In addition, Fig. 5 compares model accuracies, highlighting LR's superior performance, followed by XGBoost and 1D-CNN, with all six classifiers exceeding 99% accuracy.

Logistic Regression Accuracy: 99.79%

Classification Report:				
	precision	recall	f1-score	support
0	0.99	1.00	0.99	99
1	1.00	1.00	1.00	93
2	1.00	1.00	1.00	90
3	1.00	1.00	1.00	95
4	1.00	0.99	0.99	97
accuracy			1.00	474
macro avg	1.00	1.00	1.00	474
weighted avg	1.00	1.00	1.00	474

Fig. 4. Classification report of LR for PQD classes. Class 0: Normal, Class 1: Sag, Class 2: Swell, Class 3: Transient, Class 4: Harmonics.

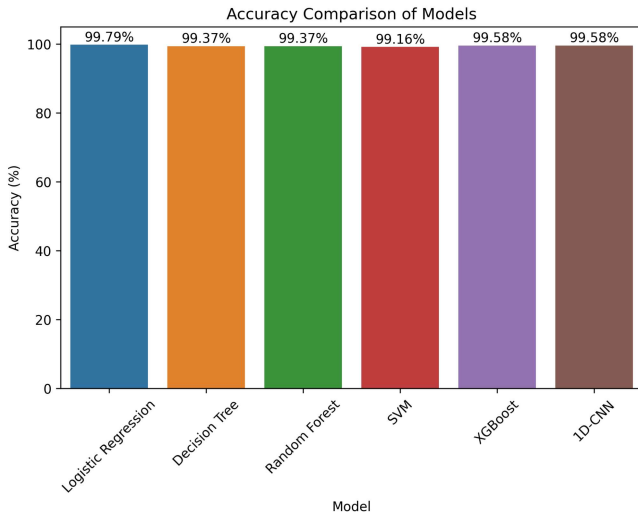


Fig. 5. Accuracy comparison of ML and DL models.

B. Cross-Validation Performance

To assess model stability and generalization, 5-fold stratified cross-validation was performed. As shown in Table V, LR obtained the highest mean accuracy of $99.45\% \pm 0.32$, with XGBoost and RF achieving $99.41\% \pm 0.49$ and $99.37\% \pm 0.30$, respectively. These findings demonstrate that the models generalize effectively and do not overfit the training data.

TABLE V
CROSS-VALIDATION PERFORMANCE OF ML AND DL MODELS.

Model	Mean Accuracy (%)
LR	99.45 ± 0.32
XGBoost	99.41 ± 0.49
RF	99.37 ± 0.30
DT	99.21 ± 0.41
SVM	99.14 ± 0.37
1D-CNN	99.38 ± 0.44

C. Confusion Matrix Analysis

Since LR achieved the highest overall accuracy, its confusion matrix is shown in Fig. 6. The matrix shows almost perfect classification, with only a single misclassification between class 4 and class 0. This highlights the strong discriminative ability of features such as THD, RMS Voltage, and Peak Voltage in distinguishing disturbance categories.

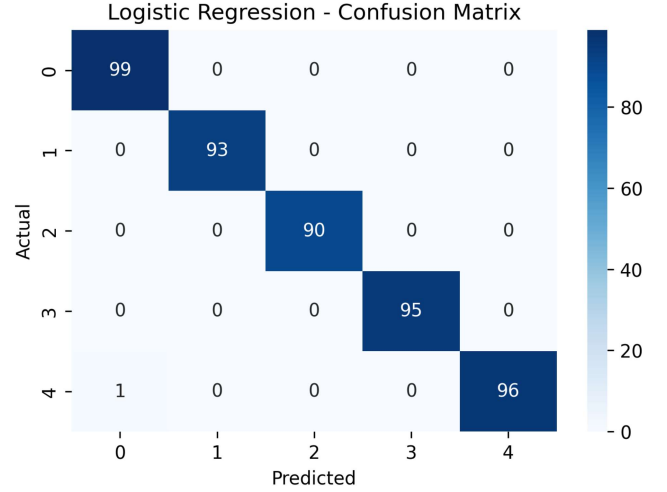


Fig. 6. Confusion matrix for LR.

D. SHAP Analysis

To interpret the decision-making process of the models, SHAP values were computed. The analysis revealed that THD, RMS Voltage, and Peak Voltage were consistently the most influential features across all models. SHAP plots provide both global and local interpretability, emphasizing the influence of individual features on model predictions across various disturbance classes, as illustrated in Fig. 7.

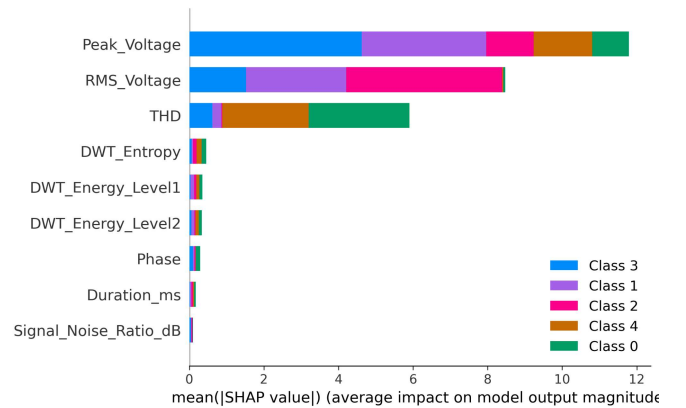


Fig. 7. SHAP summary plot showing feature importance across PQD classes.

E. Feature Importance Comparison

Tree-based feature importance was compared with SHAP results, both consistently highlighting THD, RMS Voltage, and

Peak Voltage as key features. The feature ranking comparison is shown in Fig. 8, which reinforces the interpretability of the proposed framework.

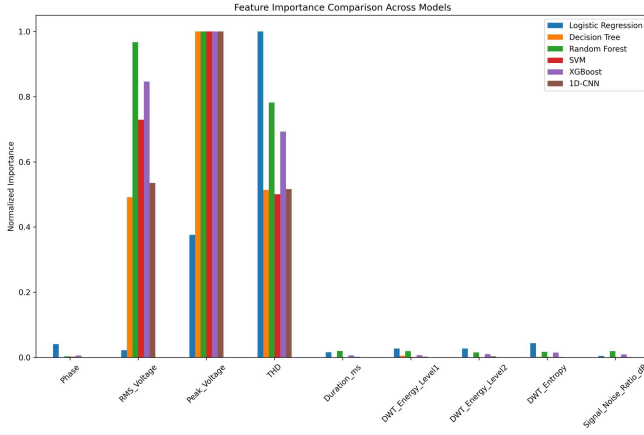


Fig. 8. Comparison of feature importance rankings across all models.

V. CONCLUSION

This paper presented an explainable ML framework for multi-class PQD classification, integrating classical algorithms and a 1D-CNN. The proposed approach combined systematic preprocessing, high-performing classifiers, and model-agnostic interpretability tools to ensure both predictive accuracy and transparency. Experimental results demonstrated that LR achieved the highest accuracy of 99.79%, while XGBoost and 1D-CNN closely followed with 99.58%, confirming the robustness of the framework. SHAP and feature-importance analyses identified critical features such as THD, RMS and peak voltage, linking predictions to physical system insights. This transparency is essential for practical deployment in smart-grid environments, where operator trust and interpretability are as important as accuracy. The models exhibit low computational overhead, making real-time deployment feasible with proper data acquisition and preprocessing. Future work includes larger datasets, handling complex hybrid disturbances, and integration into online monitoring for improved fault diagnosis and grid reliability.

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